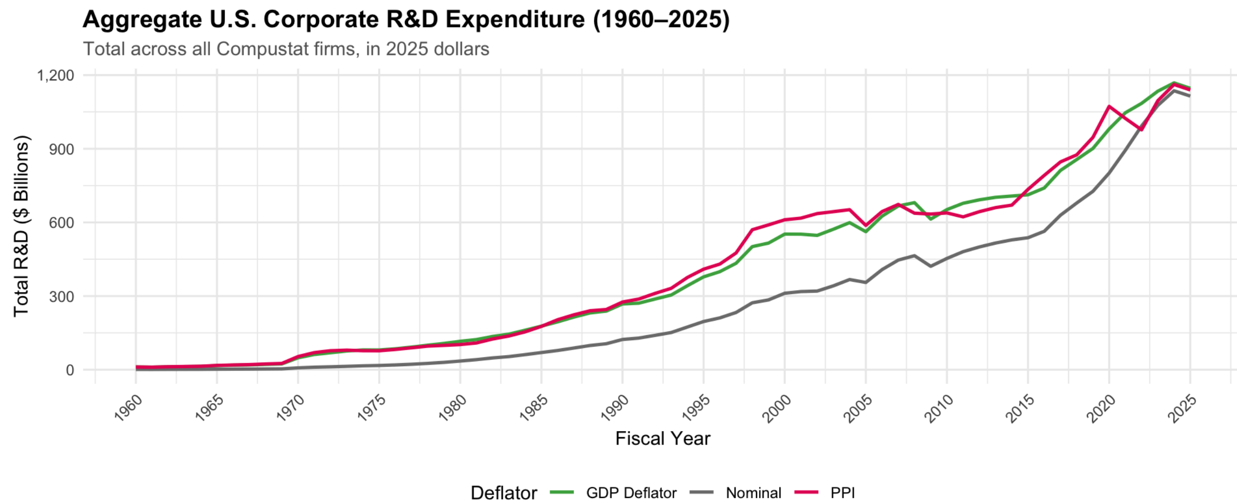
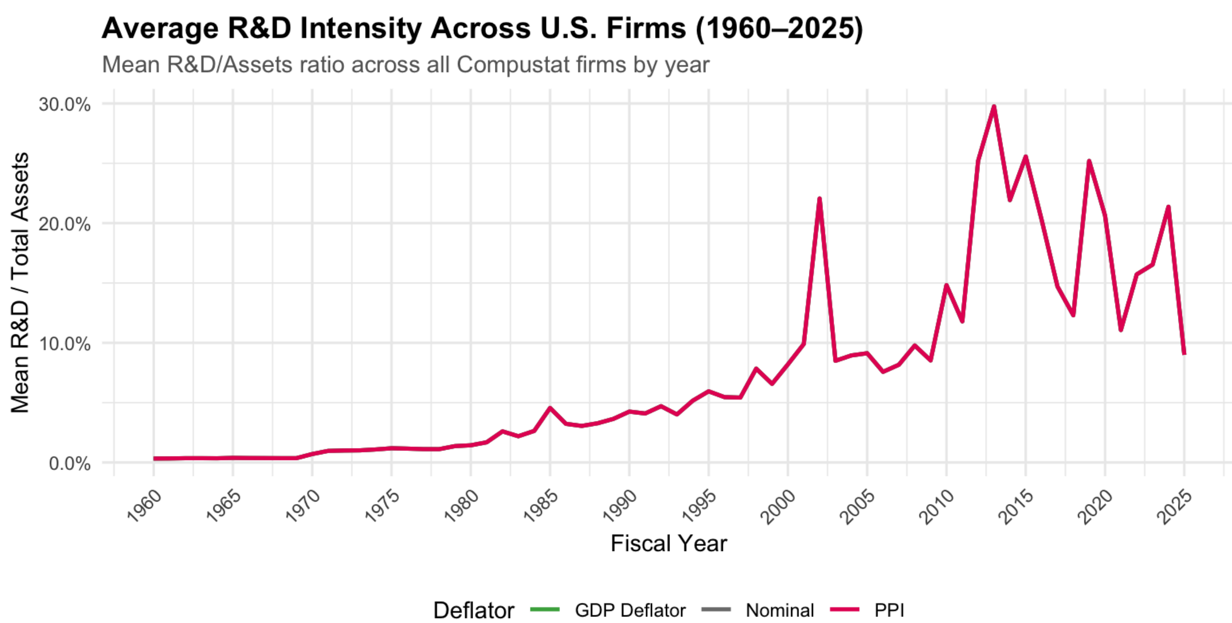
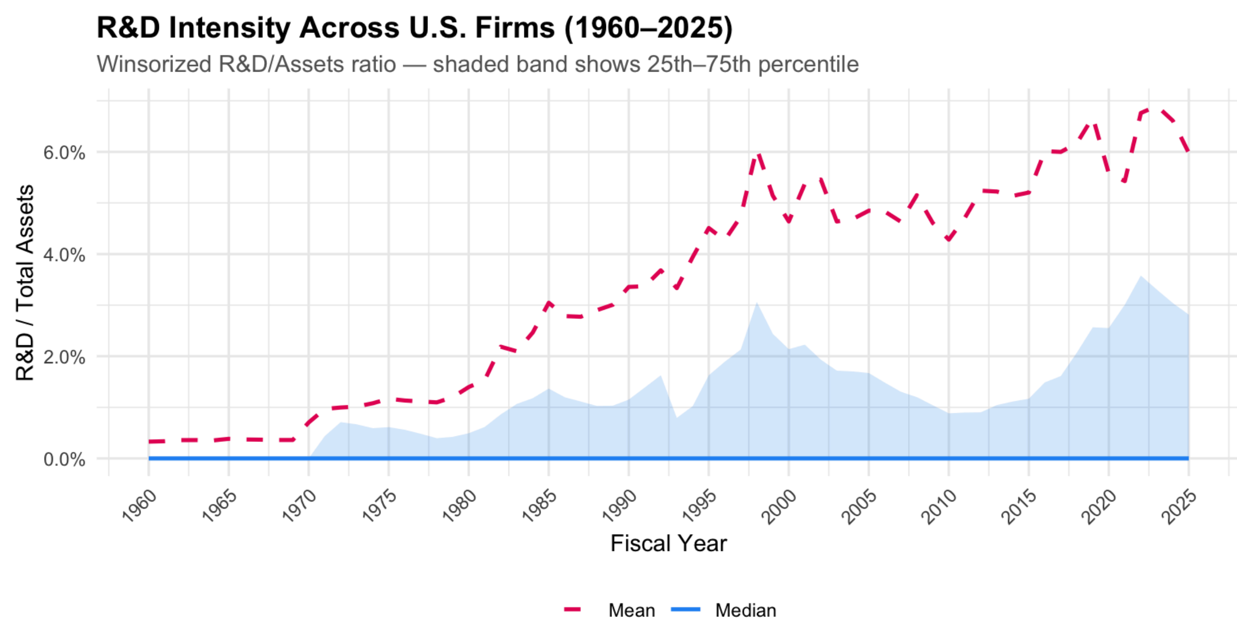


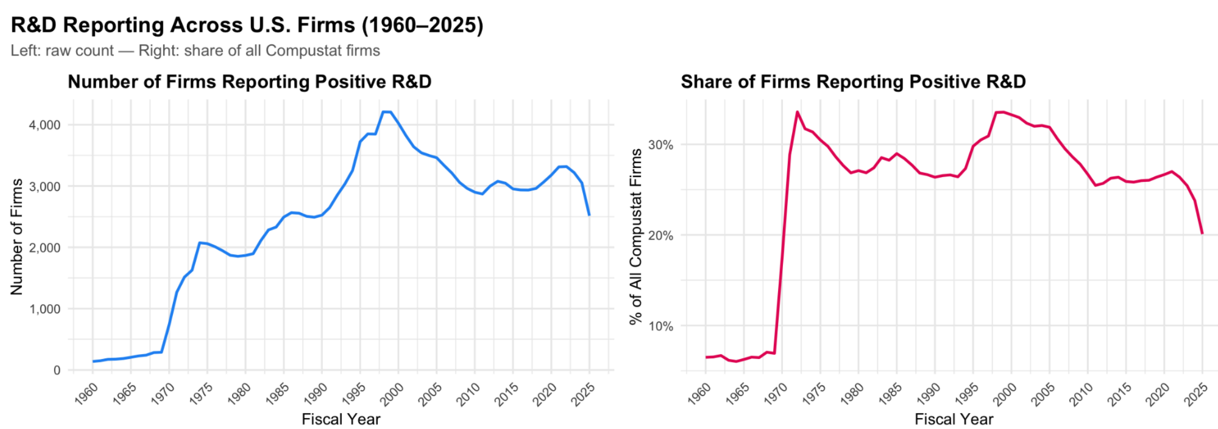


- The `is.finite()` filter removes cases where assets are 0 or missing, which would produce `Inf` or `NaN` ratios
- `rd_ratio_dist` contains all four distributional parameters (p25, median, mean, p75) for all three deflators
- The R&D/Assets ratio is computed using nominal values, as inflation adjustments cancel identically in the numerator and denominator

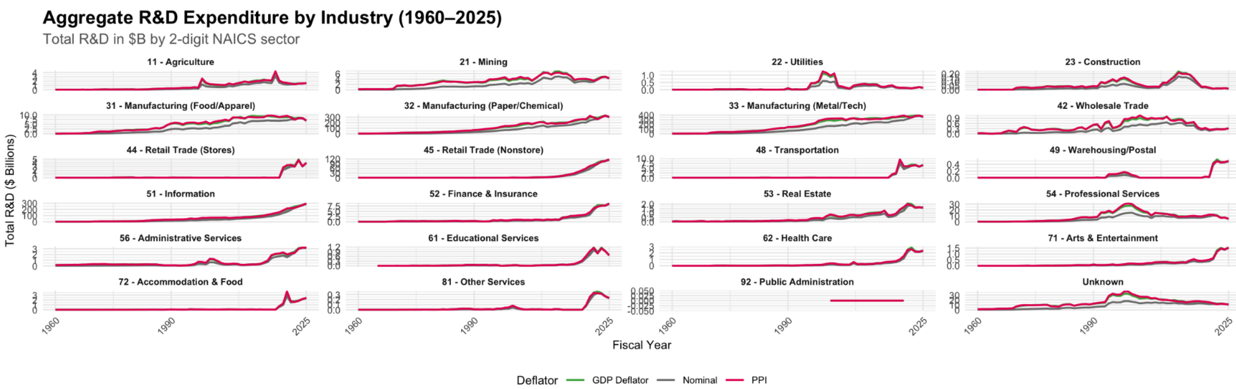


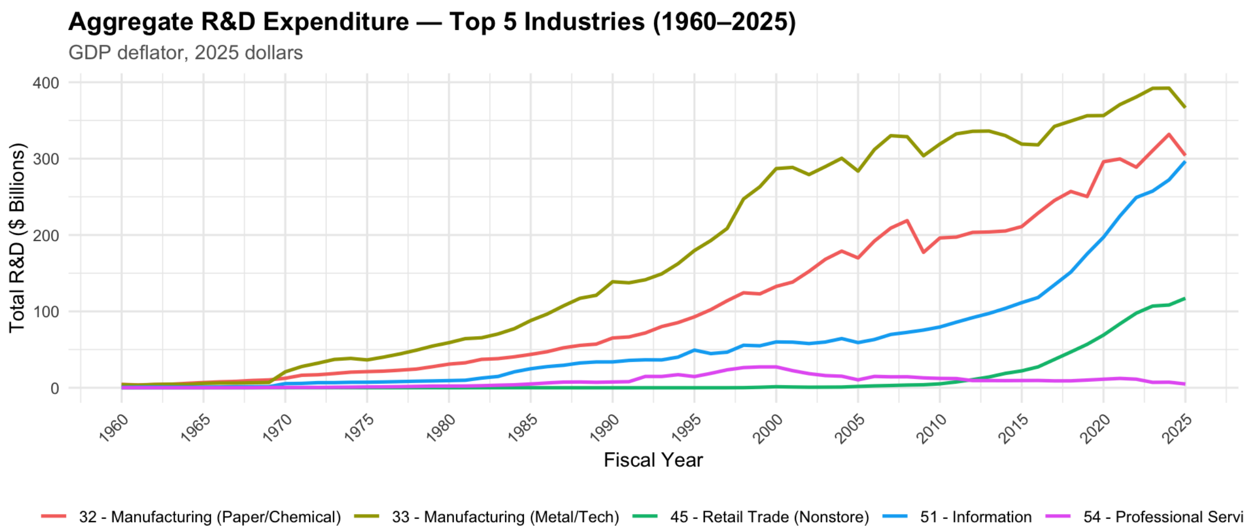
Winsorized:

- The median is flat at ~0% This means more than half of all Compustat firms report zero or near-zero R&D in every year.
- **The mean (dashed pink line) is much higher and rising.** The mean is being pulled up by a relatively small number of firms with very high R&D intensity.
- The 25th percentile is essentially 0% the entire time, and the 75th percentile only starts separating meaningfully after 2000. This confirms that **R&D reporting is skewed**, concentrated in a minority of firms.



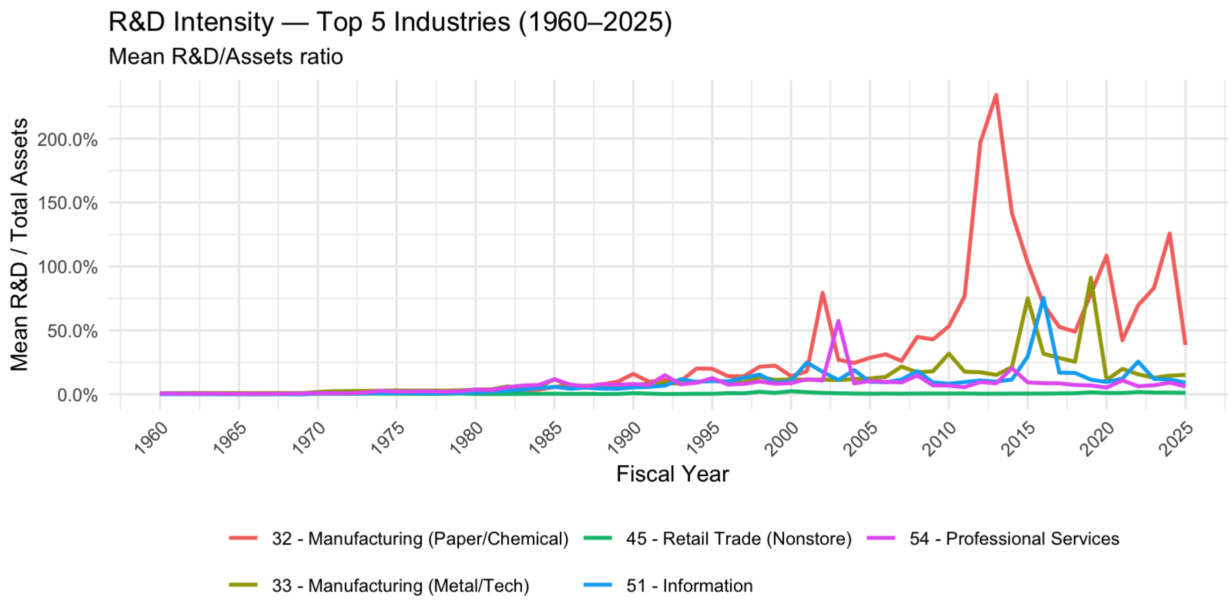
- Percentage: controlled for the fact that compustat add more firms every year
- The big jump in raw counts after 1970 on the left: Compustat expanding coverage
    - Better data from 1970?
- After 2000: Compustat added many non-R&D industries (retail, finance, services) to its coverage.

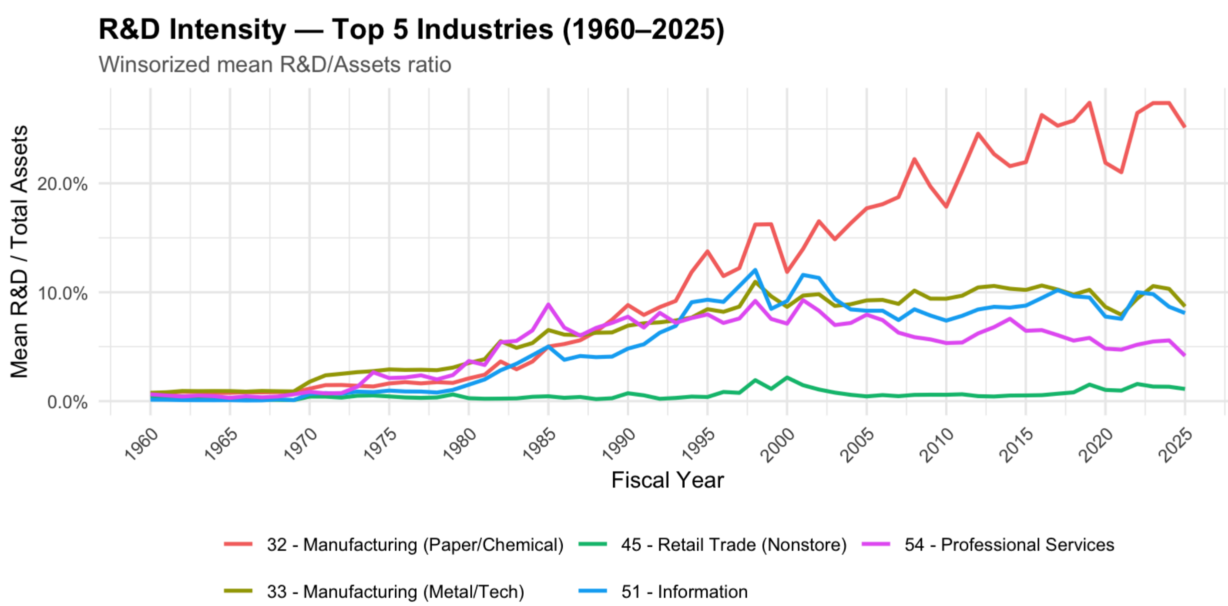




- Amazon has been coded 455219 ("Other Nonstore Retailers") - key driver of that line^

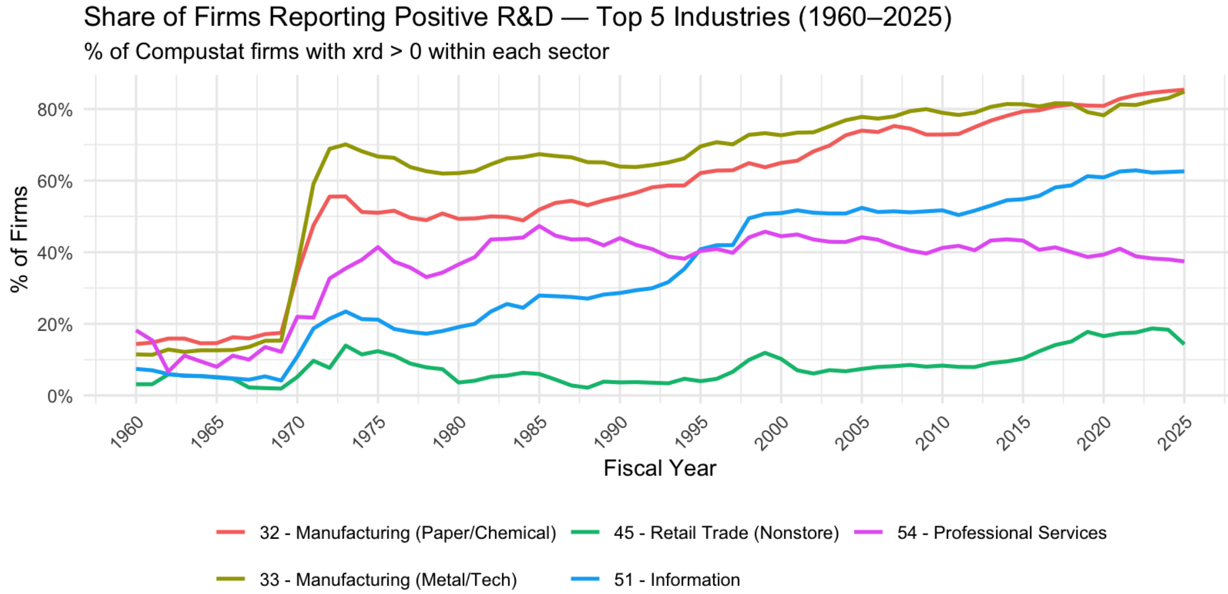
Task 5





32 - Manufacturing (Paper/Chemical) has the highest R&D intensity, rising steadily to ~25% — driven by pharma
33 - Manufacturing (Metal/Tech) and **51 - Information** track closely at ~10%, both growing
54 - Professional Services peaks around 2000 (dot-com) then declines
45 - Retail Trade (Nonstore) stays flat near 0% — confirms Amazon's R&D is large in dollar terms but tiny relative to its massive asset base

Task 6


Share of Firms Reporting Positive R&D — Top 5 Industries (1960–2025)
% of Compustat firms with xrd > 0 within each sector
80%
60%
40%
20%
0%
% of Firms
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Fiscal Year
32 - Manufacturing (Paper/Chemical)
45 - Retail Trade (Nonstore)
54 - Professional Services
33 - Manufacturing (Metal/Tech)
51 - Information